You must not read the question paper before the exam starts.

You must not write anything until the invigilator says that you may start.

Pay attention to the instructions given by the invigilators.

Read the instructions below **carefully** before the exam starts, **check** you know how you are required to complete your answer booklets or sheets.

You must ensure that the front of your answer books/sheets have been completed correctly **by the end of your exam finish time,** no further time will be given.

January 2026

EC114 2025/6 A 800

Answer Booklet
Student provided non-programmable, non-communication enabled, and non-networked calculator permitted
Graph Paper
Dictionaries not permitted

University of Reading

Introductory Macroeconomics (EC114)

Three hours

Answer **three** questions as follows
Section A: Answer **all** questions
Section B: Answer **one** question
Section C: Answer **one** question
Answers should include brief, concise and to the point explanations.
All graphs must be clearly and accurately labelled.

Section A

Answer Question 1.

1. Scarcity and Choice

(a) Briefly explain using words, graphs, and equations (if the latter exist) the following concepts.

(i) Marginal Rate of Transformation

(ii) Absolute Advantage

Indicate whether the following statement is true, false, or uncertain. Briefly explain your answer.

(iii) If the government increases the benefits for unemployed workers, resulting in more workers being unemployed and remaining unemployed for longer periods, then the PPF for the country will shift inwards.

(15 marks)

(b) A nation with fixed quantities of resources is able to produce any of the following combinations of carpet and carpet looms (needed to make carpets). Assume there exists a fixed number of looms to start with (and hence to make carpets).

Yards of Carpet (millions)	Carpet Looms (thousands)
0	45
12	42
24	36
36	27
48	15
60	0

(i) Draw the PPF for this country. Does the principle of "increasing opportunity cost" hold for this country? Does the idea of scarcity hold for this country? Briefly explain your answers.

Question continued overleaf.

(ii) If this country chooses to produce both goods, what will happen to the PPF over time? Briefly explain your answer.

(iii) Suppose that a new technology is developed that allows an additional 50 percent of yards of carpet to be produced by each existing loom. Illustrate on your graph from part (i) the impact of this new technology on the PPF.

(iv) Suppose that before the new technology was developed, the country produced 15 thousand looms. After the new technology, the country produced 27 thousand looms. What is the effect of the new technology on the production of carpet (be precise). Illustrate the before and after on your PPF graph.

(35 marks)

Section B

Answer **one** of the two questions.

2. Labour and Unemployment

(a) Briefly explain using words, graphs, and equations (if the latter exist) the following concepts.

(i) Natural Rate of Unemployment

(ii) NAIRU

Indicate whether the following statement is true, false, or uncertain. Briefly explain your answer.

(iii) An increase in worker productivity will cause the labour supply curve to shift.

(15 marks)

(b) Consider a simple economy where labour markets are characterized by the following equations for supply and demand respectively (where W is the real wage):

$$L_S = 10W - 18$$
$$L_D = 190 - 6W$$

(i) If we have a classical economy, calculate all equilibrium values for this labour market. Graphically illustrate equilibrium. How much unemployment do we have in this economy?

Now suppose instead we have a Keynesian economy. In addition, suppose that the labour force is composed of 120 individuals and that the natural rate of unemployment is 10%.

(ii) Suppose the real wage is sticky at a level below the equilibrium real wage from part (i). What can we conclude about the demand for labour? What do we know about the output gap and where we are in the business cycle? Using an AS-AD graph illustrate short-run equilibrium.

Question continued overleaf.

(iii) Using the short-run Phillips curve, what can you conclude about inflation. Show this graphically. What does the long-run Phillips indicate will happen to both inflation and unemployment? Illustrate this graphically.

(35 marks)

3. Money and Monetary Policy

(a) Briefly explain the following concepts; use graphs and equations if necessary.

(i) Money Multiplier

(ii) Open Market Operations

Indicate whether the following statement is true, false, or uncertain. Briefly explain your answer.

(iii) To increase the money supply, the Fed will purchase government securities.

(15 marks)

(b) Assume that Fleet Bank is the only bank in the economy of Readingland. No one in Readingland holds any currency (they carry out all transactions using debit cards or credit cards). Fleet Bank currently has £100 million in deposits, and faces a reserve requirement of 10%.

(i) Show a T-account for Fleet Bank. Now suppose you win £10 million in the lottery and deposit this entire amount into your current account. What is the immediate impact on reserves for Fleet Bank? How much are excess reserves?

(ii) Suppose that Fleet Bank always lends out any excess reserves. If the reserve requirement is 10%, what will be total change in deposits in this economy? Draw a T-account showing the final balance sheet of Fleet Bank.

Question continued over leaf.

(iii) Now suppose that the Bank of England announces that the new reserve requirement is 20%. What action must Fleet Bank take in order to meet the new reserve requirement? (Explain in words – no need for calculations.) What do you think the Bank of England was trying to achieve when it raised the reserve requirement?

(35 marks)

Section C

Answer **one** of the two questions.

4. Keynesian IS-LM Model

(a) Briefly explain using words, graphs, and equations (if the latter exist) the following concepts.

(i) Budget Deficit

(ii) Planned Aggregate Expenditure

Indicate whether the following statement is true, false, or uncertain. Briefly explain your answer.

(iii) An increase in the interest elasticity of investment would cause the PAE curve to shift down.

(15 marks)

(b) Consider an economy where consumption is given by $C = 100 + (1/3)Y_D$. The investment function is $I = 800 - 100r$. Government taxes are set at 400 and there is a budget surplus of 100. The demand for money in this economy depends only on the level of income; $(M/P)^D = 0.5Y + 300$. The nominal money supply is 1600, and the average price level is 2.

(i) Derive the equation for both the IS curve and the LM curve? What are the slopes of the two curves?

Question continued overleaf.

(ii) Calculate all equilibrium values for this economy (Be Precise). Illustrate equilibrium.

(iii) If government ran a balanced budget, what would happen to equilibrium output and interest rates (you need only tell me the direction of change and not the exact amount)? Briefly explain your answer and how the government could achieve this outcome.

(iv) Derive the AD curve (Be precise). Briefly explain the impact that the fiscal policy in part (iii) has on the AD curve.

(35 marks total)

5. Long-Run Keynesian Model

(a) Briefly explain using words, graphs, and equations (if the latter exist) the following concepts.

(i) Potential GDP

(ii) Demand Pull Inflation

Indicate whether the following statement is true, false, or uncertain. Briefly explain your answer.

(iii) The objective of an expansionary fiscal policy is to reduce inflation.

(15 marks)

(b) Consider an economy which is currently in both short and long-run equilibrium. As a result of Covid-19 lockdowns, suppose that AD and LRAS decreased by the same amount.

(i) In the absence of monetary or fiscal policy, briefly explain the impact on short-run equilibrium – as compared to our starting point. Illustrate these changes graphically.

Question continued over leaf.

(ii) In the absence of any policy, explain what happens to long-run equilibrium as compared to our starting point. Briefly explain (and illustrate graphically) any transition from the short run to the long run (if these differ).

(iii) After the initial short run changes as described in part (i), what actions should either the monetary or fiscal authority take to hasten our transition to long-run equilibrium? Would government action provide any benefits relative to waiting for the economy to naturally return to long-run equilibrium? Illustrate your arguments graphically.

(35 marks)

(End of Question Paper)